

Retrieval-Augmented Generation Chunking Strategies for Financial Document Analysis: A Systematic Literature Review

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Abstract—Retrieval-Augmented Generation (RAG) is an important architecture for generative AI. In particular, the subtle processing of financial reports (which constitutes a key area of regulatory scrutiny for tax authorities and financial bank sectors) has been automated. The performance of any RAG, however, is ultimately determined by its document splitting policy. When traditional fixed-size text chunking approaches are applied to financial statements, which are semi-structured, mixed data type documents, they cause context fragmentation and semantic incoherence, resulting in major degradation in subsequent retrieval and generation. This review of the literature is a systematic investigation of the emergent importance of chunking strategies within the financial context. In line with PRISMA 2020, a total of 16 recent studies were integrated in this review to unravel the effects of different chunking approaches, ranging from naive text splitting to more complex, layout-aware parsing. Our analysis shows that Naive fixed-size chunking is the most common strategy, whereas structure-aware and semantic chunking approaches enhance precision up to 33.42%. We discuss known trade-offs of chunk size granularity, retrieval precision and recall, and identify architectural patterns for high-end structure-aware solutions. The results demonstrate a conclusive and essential convergence of the previously arbitrary chunking paradigm to semantic chunking that captures, if not the entirety, a significant portion of the relational and logical nature of the document.

Keywords—Chunking, Text Segmentation, Retrieval-Augmented Generation, Large Language Models, Financial Report Analysis, Semantic Chunking, Systematic Literature Review

I. INTRODUCTION

Financial statements are complex, semi-structured artifacts blending quantitative tables with qualitative narratives [1], [2]. Since narratives provide essential context for interpreting numbers, manual synthesis is labor-intensive and error prone [3]. Research indicates that increased textual complexity raises analyst forecast dispersion [4]. While LLMs like

BloombergGPT [5] and FinGPT [6] offer automation, they often fail in text-grounded numerical reasoning when data is scattered [7], [8]. Retrieval-Augmented Generation (RAG) overcomes these limitations [9], [10], yet its efficacy depends critically on the document chunking strategy [11]. While reviews exist for other domains [12], no SLR specifically evaluates chunking in finance. This study fills that gap by mapping strategies and analyzing the precision-recall trade-off to guide high-performance financial RAG construction.

II. RESEARCH METHODOLOGY

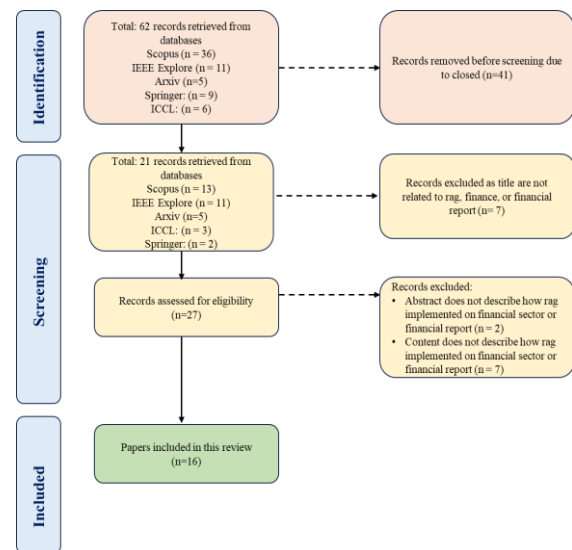


Fig. 1 The PRISMA 2020 flow diagram of the systematic literature review

This study employs a Systematic Literature Review (SLR) methodology, conducted in accordance with the PRISMA 2020 guidelines [13], to ensure a transparent and reproducible synthesis of existing research. The review is guided by the following Research Questions (RQs):

RQ1: What are the documented chunking strategies for complex financial documents, and how does the literature characterize their influence on the trade-off between retrieval precision and recall?

RQ2: What architectural patterns for data indexing and retrieval (e.g., vector database schemas, knowledge graphs, alternative search algorithms) are reported in the literature to be effective for querying semantically chunked financial data?

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To address these questions, we performed a comprehensive search across three major academic databases: Scopus, IEEE Xplore, and SpringerLink. We limited the search to literature published between January 2020 and December 2025 to capture the most recent advancements in this field. Search queries were constructed using Boolean operators to combine keywords related to the core concepts. The query string used was ('Retrieval Augmented Generation' OR 'RAG') AND ('chunking' OR 'text segmentation') AND ('financial report' OR 'finance' OR 'annual report').

To ensure a rigorous and reproducible selection process, we established formal criteria prior to screening. Inclusion criteria required studies published between January 2020 and December 2025 that primarily focused on Retrieval-Augmented Generation (RAG) applied to the financial domain, specifically analyzing financial reports, and provided explicit details on the document chunking or text segmentation strategy used. Exclusion criteria rejected records that were inaccessible (closed access), non-primary research (reviews, editorials), or irrelevant to the core intersection of RAG, chunking, and finance. Furthermore, full texts lacking sufficient technical detail regarding the RAG implementation were excluded.

The study selection followed a multi stage screening protocol, visualized in the PRISMA flow diagram. The process began with an initial search that yielded 67 records retrieved from five databases: Scopus (36), IEEE Xplore (11), Arxiv (5), Springer (9), and ICCL (6). During the identification phase, 35 records were removed before screening due to being closed access. This left 32 records for the screening phase. From this set, 7 records were excluded as their titles were not related to RAG, finance, or financial reports. Subsequently, 25 records were assessed for eligibility. From those assessed, a further 9 records were excluded because the abstract did not describe the RAG implementation (2 records) or the main content did not detail its implementation on the financial sector or reports (7 records). This rigorous process resulted in a final set of 16 qualifying studies included in the narrative synthesis. These data points included the authors and publication year, the data source used, the LLM architecture employed, the specific chunking strategy implemented, the indexing and retrieval architecture, and the reported quantitative results.

III. REVIEW RESULTS & DISCUSSION

This section presents the synthesized findings from the 16 qualifying studies included in this review. We organized the results based on the characteristics of the studies, structured to directly address the research questions concerning chunking strategies, indexing and retrieval patterns, and the resulting impact on performance. To provide a structured overview of the included studies, the main characteristics of each analyzed paper are summarized in Table 1. This table maps the various approaches, data sources, and quantitative findings that form the basis of the analysis in this review.

TABLE I
MAIN CHARACTERISTICS OF ANALYZED PAPERS

Author & Year	Chunking Strategy	Index/Retrieval Method	Key Findings
Iaroshev et al. (2024) [11]	Fixed-Size (100 tokens, 50-token overlap)	Vector DB (Weaviate)	An example of a fundamental fixed-size chunking implementation.
Gijre & Laddha (2024) [16]	Fixed-Size (1000 characters, 400-char overlap)	Hybrid (Knowledge Graph + Vector Embeddings)	The hybrid architecture achieved near-perfect scores (Faithfulness 1.0, Context Recall 0.999), ensuring fact-based answers.
Islam, Kannappan, et al. (2023) [17]	Fixed-Size (1024 tokens, 256-token overlap)	Standard Dense Retriever	Highlighted retrieval failure, with the correct answer missing from the top 5 retrieved documents in 61.3% of cases.
Hui et al. (2024) [10]	Recursive Splitting (3000-character segments)	Vector DB (text-embedding-3-large)	An example of a common recursive splitting implementation.
Mehul K et al. (2025) [18]	Recursive Splitting (LangChain's RecursiveCharacterTextSplitter)	Vector DB (FastEmbed)	Achieved a high relevance score approaching 0.90.
Bronzini et al. (2024) [19] & Malandri et al. (2025) [20]	Sentence-based Splitting / Propositions	-	Enriching a model with sentence-level propositions led to an average accuracy increase of 3.8-4.3%.
Yang, B et al. (2024) [21]	Structure-Aware (parsing into "semantic blocks": paragraphs, tables, sections)	Dense Retriever	The structure-aware method (73% accuracy) significantly outperformed a naive method (42%).
Sarathi et al. (2024) [22]	Recursive Summarization (Tree-based Structure)	Tree-Organized Vector Retrieval	Recursive tree structures significantly outperform flat chunking for long-context tasks.
Lin & Jang (2025) [15]	Concept-Based (entire semantic concept as one chunk)	Keyword-based (BM25)	Increased retrieval precision by 33% compared to a traditional method.
Jadon et al. (2025) [14]	Comparative: Fixed-Size (5, 10, 15, 20 tokens)	-	Quantified the trade-off: small chunks increased precision (31-42%), while larger chunks increased recall (18%).
Stabler et al. (2025) [23]	Comparative: Semantic, Sentence, Token	-	A sentence-splitting strategy with a 512-token window achieved the best balance (highest IoU score).
Zhu, F., et al. (2021) [24]	Hybrid (Hierarchical Tables + Text Alignment)	Dense Table-Text Retriever	Hierarchical table-text alignment is essential for accurate complex numerical reasoning.

A. Chunking Strategies and Precision-Recall (RQ1)

The analysis reveals a spectrum of chunking methodologies, ranging from mechanical splitting to semantic parsing.

1. *Fixed-Size and Recursive Chunking*: The most fundamental approach is splitting based on a fixed token count (e.g., 100 tokens [11] or 1000 characters [16]). A common variant is recursive splitting, which attempts to preserve local context by respecting separators like paragraphs [10], [18]. However, direct comparisons show that recursive splitting often fails to capture the alignment

between hierarchical tables and text, which is critical for complex numerical reasoning [24].

TABLE II
SUMMARY OF CHUNKING STRATEGY TRADE-OFFS

Strategy Type	Key Advantage	Key Disadvantage	Impact on Precision & Recall
Naive (Fixed-Size, Recursive)	Simple and fast to implement.	Causes context fragmentation; ignores document structure.	Forces a direct trade-off: small chunks for precision, large chunks for recall.
Advanced (Semantic, Structure-Aware)	Preserves semantic context and document structure.	More complex to implement; can be brittle to layout changes.	Improves both precision and recall by creating coherent, meaningful chunks.

2. *Structure-Aware and Semantic Chunking*: To overcome fragmentation, advanced studies leverage document structure. Sentence-based splitting has been shown to increase accuracy by approximately 4% by preserving propositional logic [19], [20]. More sophisticated approaches parse documents into "semantic blocks" (paragraphs, tables, sections). One study demonstrated that this structure-aware method achieved 73% accuracy on a QA task, significantly outperforming a naive method which scored only 42% [21]. Similarly, recursive tree-based indexing has been shown to significantly outperform flat retrieval by preserving the hierarchical context of documents [22].

3. *The Precision-Recall Trade-off*: The literature explicitly characterizes the trade-off: smaller chunks (<10 tokens) increase precision (31–42%) by reducing noise, while larger chunks improve recall (18%) by capturing broader context [14]. The consensus suggests that structure-aware strategies are superior because they define chunks based on logical units of meaning rather than arbitrary counts, optimizing both precision and recall [23].

Collectively, these findings demonstrate that for semi-structured documents, the precision-recall trade-off cannot be solved by simply optimizing chunk size. Instead, the literature suggests that structure-aware and semantic chunking strategies are superior, because they define chunks based on logical units of meaning, inherently balancing the need for focused content (precision) with contextual completeness (recall). Another study supports this finding, concluding that a sentence-splitting strategy with a 512-token window achieved the highest Intersection-over-Union (IoU) score, indicating a good balance [23]. The retrieval failure reported in one benchmark study [17], where the correct answer was not found in the top 5 documents in 61.3% of cases, implicitly highlights how crucial an effective chunking and retrieval strategy is for ensuring adequate recall.

B. Architectural Patterns for Indexing (RQ2)

Three primary architectural patterns were identified:

1. *Vector Databases*: The dominant pattern converts text chunks into vector embeddings (e.g., using Weaviate or

FAISS) [11], [22]. This is highly effective when combined with semantic chunking strategies.

2. *Keyword-Based Retrieval*: As an alternative, one study utilized the BM25 algorithm with "Concept-Based Chunking," achieving a 33% increase in retrieval precision over traditional methods [15].

3. *Hybrid Patterns (Knowledge Graphs)*: The most sophisticated architecture combines Knowledge Graphs with vector embeddings. Instead of a standalone vector database, embeddings are stored as attributes of nodes within a graph. This approach achieved near-perfect scores for Faithfulness (1.0) and Context Recall (0.999) [16]. This architectural evolution is significant because it attempts to bridge the historical divide between purely statistical financial analysis tools [27] and broad thematic text mining approaches like topic modeling [28], integrating them into a unified retrieval system.

Finally, benchmark studies like FinBen [25] and FINANCEBENCH [17] provide crucial context. They show that even with advanced RAG techniques, a significant performance gap remains between the best models like GPT 4 and human performance. This underscores that the field is still evolving.

IV. OPPORTUNITIES & CHALLENGES

1) Data Heterogeneity and Parsing Complexity.

Parsing diverse financial PDFs remains a major hurdle due to layout heterogeneity, whereas shifting tables and variable formatting break standard parsers [20], [23]. While multimodal LLMs offer a potential solution for universal parsing, a significant gap remains between academic research and practice. Many studies rely on clean benchmarks like FinQA [7] or TAT-QA [8], creating a false sense of security. Real-world applications face "messy" scanned documents that often fail on systems optimized for sanitized datasets like FINANCEBENCH [17].

2) Preserving Semantic Integrity.

Preserving semantic integrity during chunking is critical [23], [26]. Blind fixed-size splitting often severs the link between data points—such as separating a table row from its header—triggering irreversible "error cascades" [3]. If the initial extraction is flawed, the downstream LLM generation inevitably fails.

3) Accuracy versus Efficiency

There is an inherent trade-off between accuracy and computational efficiency. While structure-aware strategies are superior, they require significant GPU resources and complex maintenance. This poses a dilemma for institutions needing to process thousands of reports, where the scalability of storage and retrieval latency becomes a critical bottleneck.

Scalability is another issue. A system that works for twelve reports might crash when handling thousands. Storage costs for vectors go up, and retrieval slows down. Plus, as reporting standards change, custom parsers become obsolete. The constant work needed to update these parsers is a hidden cost that many people overlook.

V. CONCLUSION

In this systematic review, we explored how document chunking defines the success of Retrieval Augmented Generation (RAG) systems in finance. The evidence is clear. Many factors influence a RAG system, such as the embedding model or the LLM itself. However, the chunking strategy is the foundation. It is arguably the most decisive step. If a mistake happens here, like severing a number from its context, it triggers an error cascade. No downstream component can fix this. If the right context is lost due to bad splitting, even the most powerful LLM will fail. Studies like Studies like Yang et al. [21] proved this point. They demonstrate that specialized retrieval pipelines significantly outperform naive baselines by correctly preserving the context of financial tables and narratives. This proves that chunking is not just a preprocessing task. It is a core architectural choice that sets the ceiling for the system performance. We must also admit a limitation in the current research. None of the papers we reviewed formally measured how much of the performance variance. This is a big gap. Future work needs to design controlled experiments to isolate and measure the specific impact of each part of the RAG pipeline. To answer our guiding research questions, we found three things. First, chunking strategies matter immensely. Naive methods fail because they fragment the text, while semantic methods yield massive gains. Second, the precision recall tradeoff is not solved by finding a magic number for chunk size. Instead, we need strategies that create variable chunks based on meaning. Third, the field is moving beyond simple vector search. We see hybrid architecture that combine embeddings with knowledge graphs. Future research must focus on standardizing these parsers to handle the global variability of financial reporting.

AUTHOR CONTRIBUTIONS

R.A.: Conceptualization, Methodology, Writing – original Draft Preparation, Visualization. M.M.H.: Writing - Review & Editing, Validation. A.I.: Writing - Review & Editing. B.P.: Writing – Review & Editing, Supervision.

DATA AVAILABILITY STATEMENT

The list of papers included in this systematic literature review (SLR) is publicly available at: <https://s.id/gdk2Kz>.

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